Lecture 10 - Probabilistic Machine Learning

ECE 447: Data Analysis and Machine Learning for Engineers

Bahareh Tolooshams, Ph.D.
Assistant Professor,
University of Alberta
btolooshams@ualberta.ca
https://btolooshams.github.io

February 24 and March 03, 2026

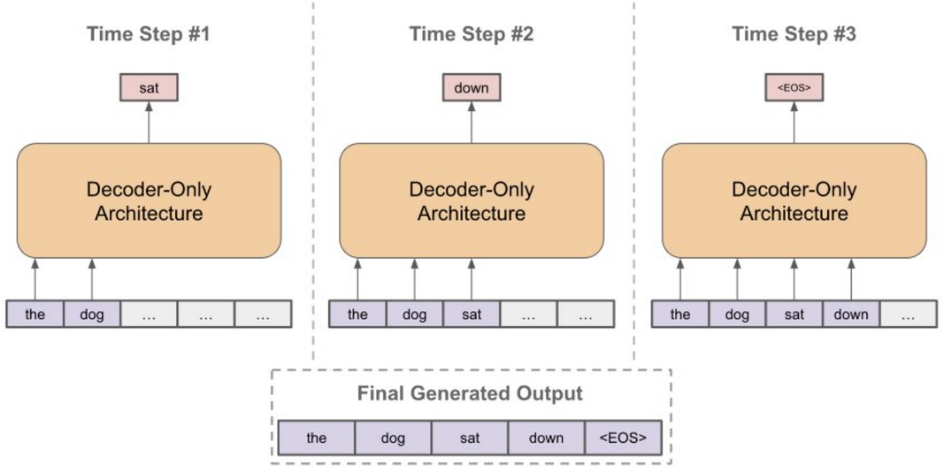
Thanks to those given us feedback.

Please take a moment to fill in the *Mid-Term Feedback Form*.

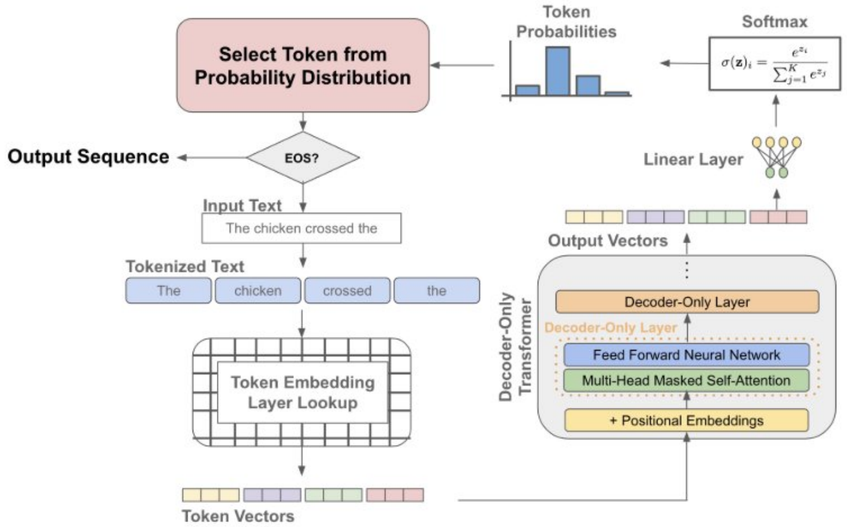
Large Language Models

This slide is informational, excluded from assessment.

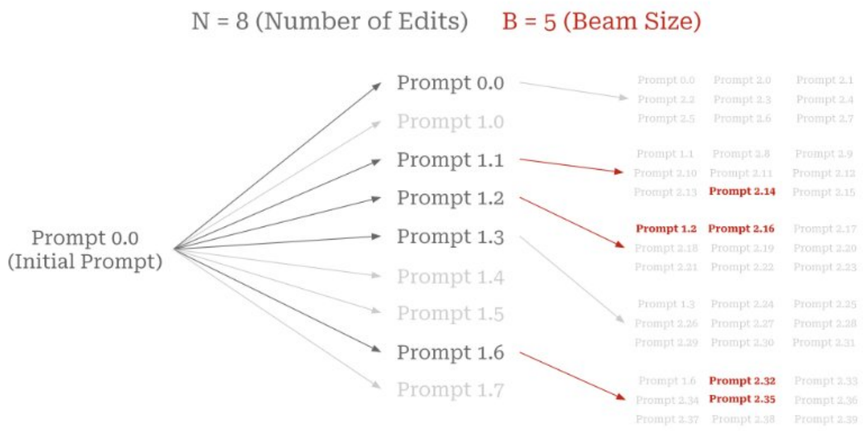
LLMs perform autoregressive next token prediction to generate text



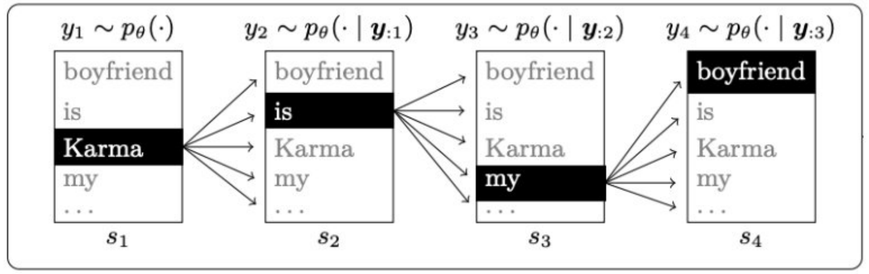
The LLM outputs a probability distribution over its vocabulary



MAP-based decoding tries to find the most probable sequence



Sampling is used by all LLM APIs and simply samples from the (transformed / truncated) token distribution predicted by the LLM



Cameron R. Wolfe, Ph.D.
@cwolferesearch
Subscribe

This Lecture Covers ...

- Probabilistic Perspectives (Maximum Likelihood)
- Re-derive Linear Regression and Weighted Least Squares
- Interpret Classifiers with Probabilistic Perspectives
- Logistic Regression for Classification
- Bayesian 2-class Classification with Maximum A Posteriori (MAP)

Independent and Identically Distributed (*i.i.d.*) Random Variables

The random variables $Y_1, Y_2, \ldots, Y_n$ are i.i.d. if they all have the same pdf/pmf.

Examples:

Gaussian distribution

$$Y_1, Y_2, \ldots, Y_n \text{ i.i.d. } \mathcal{N}(\mu, \sigma^2)$$

$$p_{Y_i}(y_i) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(y_i-\mu)^2}{\sigma^2}}$$

$$i = 1, \ldots, n$$

Bernoulli distribution

$$Y_1, Y_2, \ldots, Y_n \text{ i.i.d. } \mathrm{Bern}(p)$$

$$P(Y_i = 1) = p \qquad Y_i \in \{0,1\}$$

$$i = 1, \ldots, n$$

How to Estimate Unknown Parameters of a Distribution?

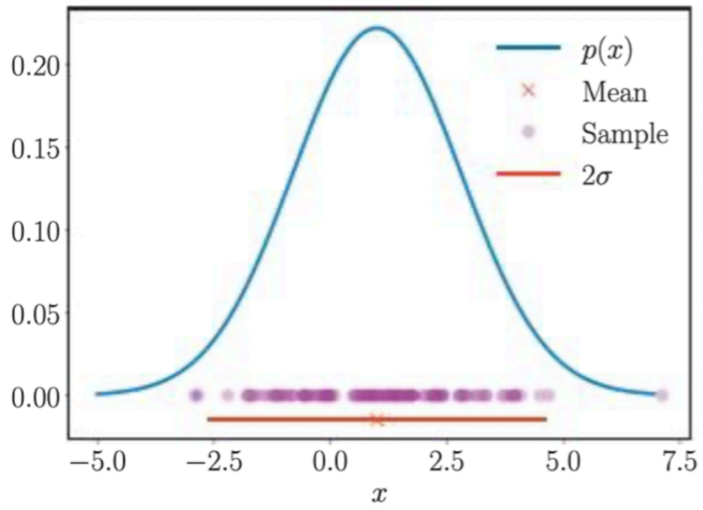
Goal: Use the available data to estimate the unknown parameters.

learn

Ex. You have collected data. You plot its histogram, and it looks like this

You model it with Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$,

and estimate the mean and variable from data.



The likelihood Function

The likelihood function of n random variables $Y_1, \ldots, Y_n$, parameterized by $\boldsymbol{\theta} \in \mathbb{R}^p$, is a join pdf/pmf of the random variables written as a function of $\boldsymbol{\theta}$

$$L(\boldsymbol{\theta} \mid y_1, \ldots, y_n) = \quad p_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n; \boldsymbol{\theta})$$

Ex. Let $Y_1, \ldots, Y_n$ be iid

$$\mathcal{N}(\mu, \sigma^2) \qquad \theta = (\mu, \sigma^2) \in \mathbb{R}^2$$

$$\mathcal{N}(\mu, 1) \qquad \theta = \mu \in \mathbb{R}$$

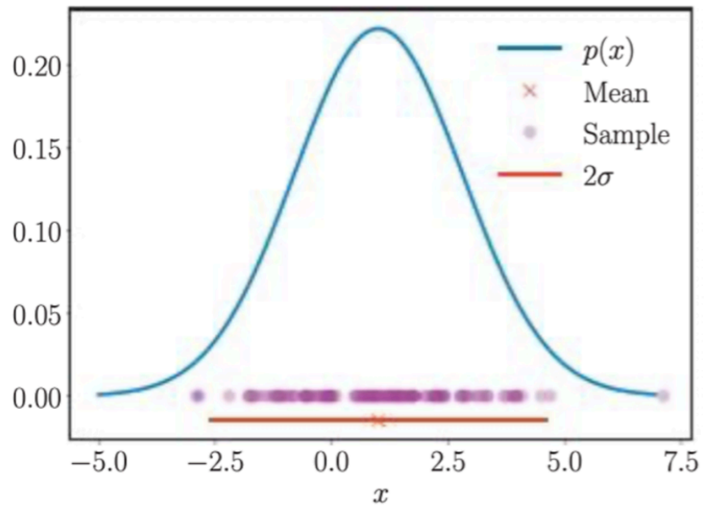
$$\text{Bern}(p) \qquad \theta = p$$

The Maximum Likelihood (ML) Estimation (informal)

Find the parameter value that best explains your data.

Given distribution $\mathcal{N}(\mu, \sigma^2)$, what is the parameter μ that best explains the data?

You compute the sample mean to estimate μ.



The Maximum Likelihood (ML) Estimation (formal)

Given a likelihood function $L(\theta \,|\, y_1, \ldots, y_n)$, the ML estimator $\hat{\theta}_{\text{ML}}$ of θ is a function of $y_1, \ldots, y_n$ that maximizes the likelihood function.

the joint probability distribution.

$$\hat{\theta}_{\text{ML}} = \arg\max_{\theta} \; L(\theta \,|\, y_1, \ldots, y_n)$$

$$= \arg\max_{\theta} \; \log \; L(\theta \,|\, y_1, \ldots, y_n)$$

Lets prove that you were doing ML estimation without knowing, when you chose the sample mean as your parameter estimate in previous slide.

Maximum Likelihood Estimate for Mean of Gaussian Distribution

Let $Y_1, Y_2, \ldots, Y_n$ be iid $\mathcal{N}(\mu, \sigma^2)$, where $\theta = \mu$. Find $\hat{\theta}_{\text{ML}}$.

$$y_i = \mu + \epsilon_i \qquad , \epsilon_i \sim \mathcal{N}(0,\sigma^2)$$

$$L(\mu \,|\, y_1, \ldots, y_n) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(\frac{-\sum_{i=1}^{n} (y_i - \mu)^2}{2\sigma^2}\right)$$

$$\hat{\theta}_{\text{ML}} = \hat{\mu}_{\text{ML}} = \arg\max_{\theta} \ \log \ L(\mu \,|\, y_1, \ldots, y_n)$$

Lets Simplify the Log Likelihood

$$\log\ L(\mu\,|\,y_1,\ldots,y_n) = \log\left(\frac{1}{(2\pi\sigma^2)^{n/2}}\exp\left(\frac{-\sum_{i=1}^{n}(y_i-\mu)^2}{2\sigma^2}\right)\right)$$

$$= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2$$

$$\hat{\theta}_{\mathsf{ML}} = \arg\max_{\theta}\ \log\ L(\mu\,|\,y_1,\ldots,y_n) = \arg\max_{\theta}\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2\right)$$

$$= \arg\max_{\theta}\left(-\sum_{i=1}^{n}(y_i-\mu)^2\right)$$

Have you seen loss this before?

$$= \arg\min_{\theta}\sum_{i=1}^{n}(y_i-\mu)^2$$

Gaussian Assumption Underlies Euclidean Loss (ℓ_2-norm)

When you ***minimize a Euclidean loss (ℓ_2-norm)****, you have placed a* ***Gaussian distributed assumption*** *on noise statistics.*

We got the following optimization objective

$$\hat{\theta}_{\mathsf{ML}} = \arg\min_{\theta} \sum_{i=1}^{n} (y_i - \mu)^2$$

for the model of data as $Y \sim \mathcal{N}(\mu, \sigma^2)$

$$y_i = \mu + \epsilon_i \qquad , \epsilon_i \sim \mathcal{N}(0,\sigma^2)$$

The noise of data is Gaussian distributed.

To find $\hat{\theta}_{\text{ML}}$, Take the Derivative Set to Zero

$$\frac{d}{d\mu} \log \; L(\mu \,|\, y_1, \ldots, y_n) = 0$$

$$\frac{d}{d\mu} \sum_{i=1}^{n} (y_i - \mu)^2 = 0$$

$$2 \sum_{i=1}^{n} (y_i - \mu) = 0$$

$$n\mu = \sum_{i=1}^{n} y_i \quad \Rightarrow$$

*Under **Gaussian** noise statistics, the estimate that **best explain your data** is sample mean.*

$$\hat{\theta}_{\text{ML}} = \hat{\mu}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^{n} y_i$$

Maximum Likelihood estimate.

What is the ML estimate for p in $\mathrm{Bern}(p)$? (I)

Let $Y_1, \ldots, Y_n$ be iid Bernoulli random variable with $P(y = 1) = p$. Denote $\theta = p$. Estimate θ such that it best describes the data, i.e. maximizes its pmf.

First, we compute the log likelihood:

$$L(p \,|\, y_1, \ldots, y_n) = \prod_{i=1}^{n} p^{y_i}(1-p)^{1-y_i}$$

$$= p^{\sum_{i=1}^{n} y_i}(1-p)^{n-\sum_{i=1}^{n} y_i}$$

$$\log L(p \,|\, y_1, \ldots, y_n) = (\sum_{i=1}^{n} y_i)\log p + (n - \sum_{i=1}^{n} y_i)\log(1-p)$$

What is the ML estimate for p in $\text{Bern}(p)$? (II)

$$\log L(p \,|\, y_1, \ldots, y_n) = (\sum_{i=1}^{n} y_i) \log p + (n - \sum_{i=1}^{n} y_i) \log(1-p)$$

$$\frac{d}{dp} \log L(p \,|\, y_1, \ldots, y_n) = 0$$

$$(\sum_{i=1}^{n} y_i) \frac{1}{p} - (n - \sum_{i=1}^{n} y_i) \frac{1}{1-p} = 0$$

$$(\sum_{i=1}^{n} y_i)(\frac{1}{p} + \frac{1}{1-p}) - n \frac{1}{1-p} = 0$$

$$(\sum_{i=1}^{n} y_i)(\frac{1}{p} + \frac{1}{1-p}) = n \frac{1}{1-p}$$

$$(\sum_{i=1}^{n} y_i)(\frac{1-p}{p} + 1) = n$$

$$(\sum_{i=1}^{n} y_i)(\frac{1}{p}) = n$$

$$\hat{p}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^{n} y_i$$

Your estimate on p to ***best explain the data*** *is "count the proportion of 1s over all 0,1s".*

Probabilistic Perspective on Linear Regression (I)

Given a set of data points $\{\boldsymbol{x}_i, y_i\}_{i=1}^n$, suppose we construct the linear function

$$y_i = \boldsymbol{\theta}^\top \boldsymbol{x}_i + \epsilon_i \qquad , \boldsymbol{x}_i \in \mathbb{R}^p \qquad i = 1,\ldots,n$$

$$, \epsilon_i \sim \mathcal{N}(0,\sigma^2) \qquad \textit{iid}$$

Let $\boldsymbol{X} = \begin{bmatrix} -\boldsymbol{x}_1^\top - \\ -\boldsymbol{x}_2^\top - \\ \vdots \\ -\boldsymbol{x}_n^\top - \end{bmatrix} \in \mathbb{R}^{n\times p} \qquad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \in \mathbb{R}^n,$

We get the familiar linear equation $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$.

Probabilistic Perspective on Linear Regression (II)

$$\boldsymbol{y} = \boldsymbol{X\theta} + \boldsymbol{\epsilon}$$

Goal: Find the maximum likelihood of your parameter $\hat{\boldsymbol{\theta}}_{\text{ML}}$.

What is the probability distribution of $\boldsymbol{y}$?

$$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{X\theta}, \sigma^2 \boldsymbol{I}) \qquad p_{\boldsymbol{Y}}(\boldsymbol{y}) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{\frac{-1}{2\sigma^2}\|\boldsymbol{y}-\boldsymbol{X\theta}\|_2^2}$$

Log likelihood: $\log L(\boldsymbol{\theta}\,|\,\boldsymbol{y}) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X\theta}\|_2^2$

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg\max_{\boldsymbol{\theta}} \log L(\boldsymbol{\theta}\,|\,\boldsymbol{y}) = \arg\max_{\boldsymbol{\theta}} -\frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X\theta}\|_2^2$$

Maximum Likelihood (ML)!

$$= \arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma^2}\|\boldsymbol{y} - \boldsymbol{X\theta}\|_2^2$$

Least Square (LS)!

Why Probabilistic Perspective?

It gives us a principled tool to account for statistics of data in modelling.

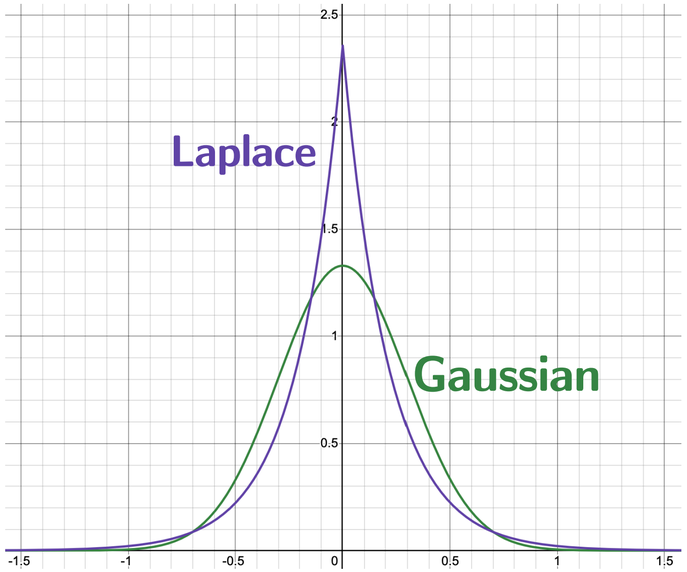
Ex1. What if you want to do linear regression, and your noise is non-Gaussian, concentrated more around zero; resembling Laplace distribution.

$$y_i = \boldsymbol{\theta}^\top \boldsymbol{x}_i + \epsilon_i \qquad , \boldsymbol{x}_i \in \mathbb{R}^p \qquad i = 1, \ldots, n$$

$$, \epsilon_i \sim \text{Laplace}(\lambda) \qquad \textit{iid}$$

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg\max_{\boldsymbol{\theta}} -\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_1$$

$$= \arg\min_{\boldsymbol{\theta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta}\|_1$$



https://www.desmos.com/calculator

What If Your Noise is Non-iid?

Consider the linear model with non-iid noise statistics.

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\theta} + \boldsymbol{\epsilon} \qquad , \boldsymbol{\epsilon} \sim \mathcal{N}(0, \boldsymbol{\Sigma})$$

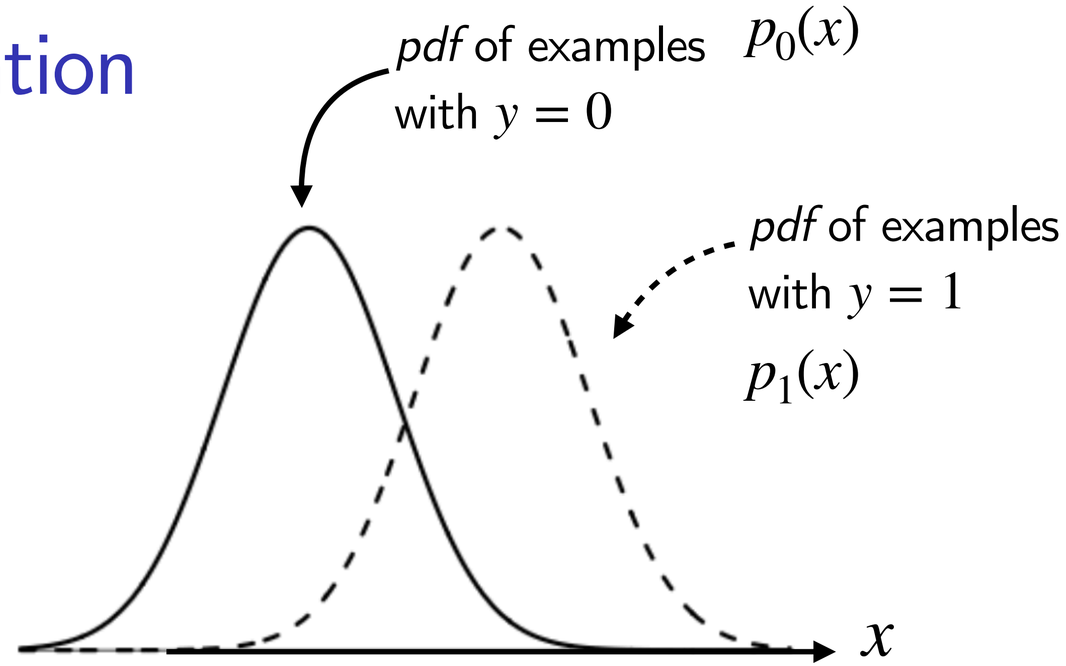
Likelihood function: $L(\boldsymbol{\theta} \,|\, \boldsymbol{y}) = \dfrac{1}{(2\pi)^{n/2} \,|\, \boldsymbol{\Sigma} \,|^{1/2}} e^{\frac{-1}{2}(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{y}-\boldsymbol{X}\boldsymbol{\theta})}$

Log likelihood: $\log L(\boldsymbol{\theta} \,|\, \boldsymbol{y}) = -\dfrac{n}{2}\log 2\pi - \dfrac{1}{2}\log|\boldsymbol{\Sigma}| - \dfrac{1}{2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})$

$$\hat{\boldsymbol{\theta}}_{\text{ML}} = \arg\max -\frac{1}{2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\theta})$$

Weighted Least Squares (WLS)

Probabilistic Perspective on Classification



Given a set of data points $\{x_i, y_i \in \{0,1\}\}_{i=1}^n$,

assign $\boldsymbol{x}_i$ to the class which it is most probable to come from.

$$x_i \in c_k \text{ iff } p_k(x) > p_l(x) \quad \text{for all } l \neq k$$

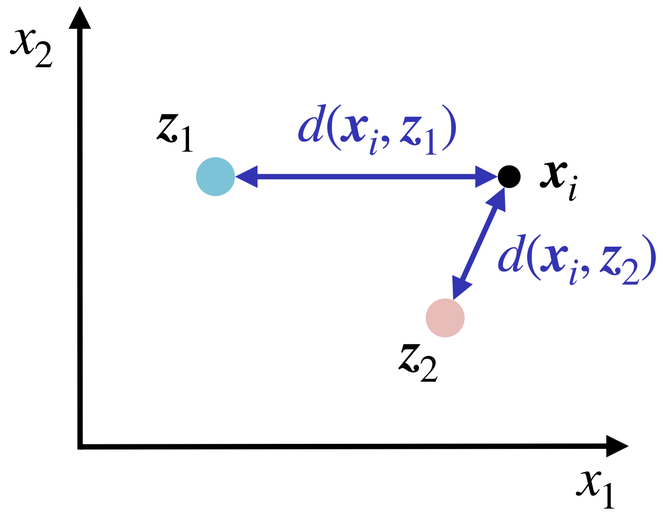
What does this remind you of?

Linear discriminant function via minimum distance classifier.

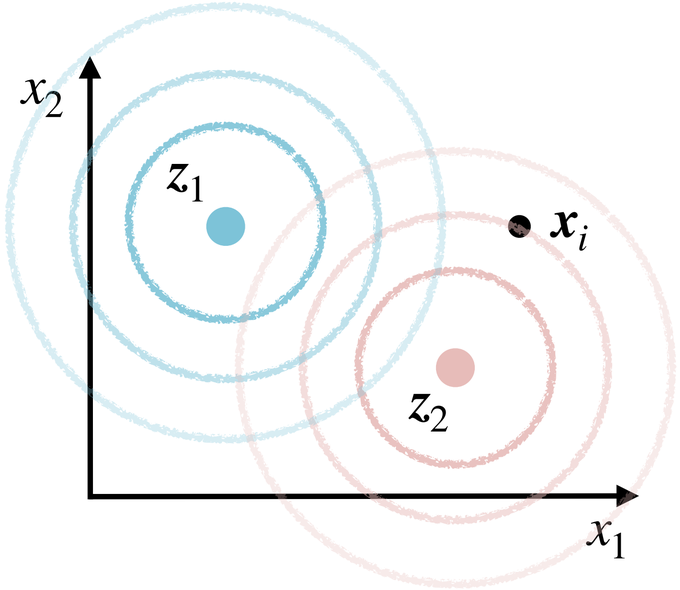
$$\boldsymbol{x}_i \in c_k \text{ iff } d(\boldsymbol{x}_i, \boldsymbol{z}_k) < d(\boldsymbol{x}_i, \boldsymbol{z}_l) \quad \text{for all } l \neq k$$

Two Perspectives on the Same Classification Method

Define a **distance metric** and assign to the class with **smallest distance** to.



Define a **probability distribution** and assign to the class with **maximum likelihood**.



Why Is This Perspective Useful?

You can derive lots of ML classifiers with only *one logic:*

*assign $\boldsymbol{x}_i$ to the **most probable class** to come from.*

Methods differ from one another in their **assumptions** to the **data distribution**.

Re-visit Minimum Euclidean Distance (MED) Classifier

Given a set of data $\{\boldsymbol{x}_i \in \mathbb{R}^p, y_i \in \{-1,1\}\}_{i=1}^n$, we define the classification decision as

$$\boldsymbol{x}_i \in c_k \text{ iff } p_k(\boldsymbol{x}) > p_l(\boldsymbol{x}) \qquad \text{for all } l \neq k$$

where $p_k(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{z}_k, \sigma^2 \boldsymbol{I})$.

$$\frac{1}{(2\pi\sigma^2)^{p/2}} e^{\frac{-1}{2\sigma^2}\|\boldsymbol{x}_i - \boldsymbol{z}_k\|_2^2} > \frac{1}{(2\pi\sigma^2)^{p/2}} e^{\frac{-1}{2\sigma^2}\|\boldsymbol{x}_i - \boldsymbol{z}_l\|_2^2}$$

(take the log) $$\frac{-1}{2\sigma^2}\|\boldsymbol{x}_i - \boldsymbol{z}_k\|_2^2 > \frac{-1}{2\sigma^2}\|\boldsymbol{x}_i - \boldsymbol{z}_l\|_2^2 \quad \Rightarrow$$

$$\boldsymbol{x}_i \in c_k \text{ iff}$$

$$\|\boldsymbol{x}_i - \boldsymbol{z}_k\|_2^2 < |\boldsymbol{x}_i - \boldsymbol{z}_l\|_2^2$$

This is **MED**. You have assumed classes are distributed as $\mathcal{N}(\boldsymbol{z}_k, \sigma^2 \boldsymbol{I})$.

Re-visit Minimum Weighted Euclidean Distance (WED) (I)

Given a set of data $\{\boldsymbol{x}_i \in \mathbb{R}^p, y_i \in \{-1,1\}\}_{i=1}^n$, we define the classification decision as

$$\boldsymbol{x}_i \in c_k \text{ iff } p_k(\boldsymbol{x}) > p_l(\boldsymbol{x}) \qquad \text{for all } l \neq k$$

where $p_k(\boldsymbol{x}) = \mathcal{N}(z_k, \boldsymbol{\Sigma})$.

We now have a general covariance matrix, same for both class

$$\frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} e^{\frac{-1}{2}(\boldsymbol{x}_i - z_k)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i - z_k)} > \frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} e^{\frac{-1}{2}(\boldsymbol{x}_i - z_l)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i - z_l)}$$

$$\frac{-1}{2}(\boldsymbol{x}_i - z_k)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i - z_k) > \frac{-1}{2}(\boldsymbol{x}_i - z_l)^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_i - z_l)$$

Re-visit Minimum Weighted Euclidean Distance (WED) (II)

$$(\boldsymbol{x}_i - \boldsymbol{z}_k)^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}_i - \boldsymbol{z}_k) < (\boldsymbol{x}_i - \boldsymbol{z}_l)^\top \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}_i - \boldsymbol{z}_l)$$

If you can decompose $\boldsymbol{\Sigma}^{-1}$ in terms of $\boldsymbol{W}^\top \boldsymbol{W}$, then we get

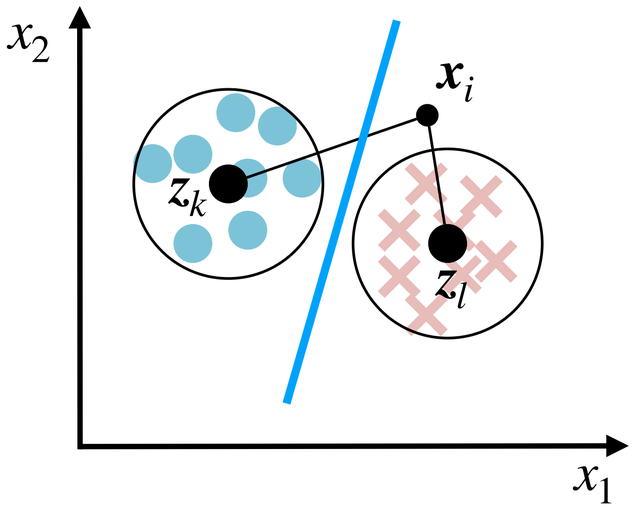
$$\|\boldsymbol{W}(\boldsymbol{x}_i - \boldsymbol{z}_k)\|_2^2 < \|\boldsymbol{W}(\boldsymbol{x}_i - \boldsymbol{z}_l)\|_2^2$$

$$\boldsymbol{x}_i \in c_k \text{ iff } \|\boldsymbol{W}(\boldsymbol{x}_i - \boldsymbol{z}_k)\|_2^2 < \|\boldsymbol{W}(\boldsymbol{x}_i - \boldsymbol{z}_l)\|_2^2 \quad \text{for all } l \neq k$$

This is **WED**. You have assumed classes are distributed as $\mathcal{N}(\boldsymbol{z}_k, \boldsymbol{\Sigma})$.

Underlying Assumptions behind MED and WED

Minimum Euclidean Distance

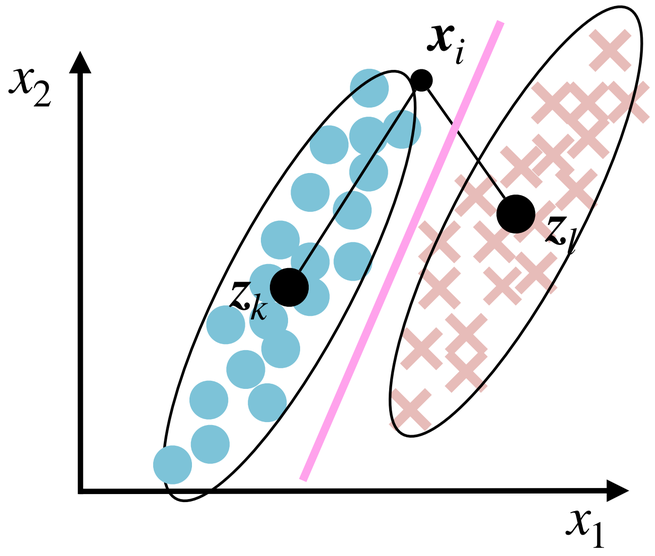


Features are not much correlated

The classes/features have similar unit/scale/variance

$$p_k(\boldsymbol{x}) = \mathcal{N}(z_k, \sigma^2 \boldsymbol{I})$$

Minimum Weighted Euclidean Distance



Features are correlated

The features have different unit/scale/variance

$$p_k(\boldsymbol{x}) = \mathcal{N}(z_k, \boldsymbol{\Sigma})$$

What if the Covariance of Data from Each Class Differs?

Your classifier won't be *linear* anymore!

Lets discuss a few examples on the board.

You might see this in future assignments!

Logistic Regression — Setup

Let $\{y_i \in \{0,1\}, \boldsymbol{x}_i \in \mathbb{R}^p\}_{i=1}^n$ and $\boldsymbol{\theta} \in \mathbb{R}^p$. For a given $\boldsymbol{x}_i$,

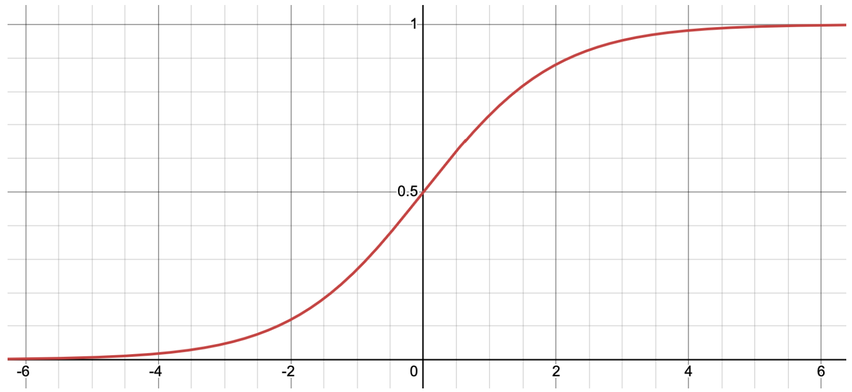
$$y_i \mid \boldsymbol{x}_i \sim \mathrm{Bern}(\pi_i) \qquad y_i = \begin{cases} 1, & \text{with probability } \pi_i \\ 0, & \text{with probability } (1-\pi_i) \end{cases}$$

Goal: Estimate π_i.

Construct a linear function $\boldsymbol{\theta}^\top \boldsymbol{x}_i$ and then map it to a prediction score between 0,1.

$$\pi_i = \sigma(\boldsymbol{\theta}^\top \boldsymbol{x}_i)$$

sigmoid function $\sigma(z) = \dfrac{1}{1+e^{-z}}$



Logistic Regression — The Sigmoid Function

$$\pi_i = \sigma(\boldsymbol{\theta}^\top \boldsymbol{x}_i) = \frac{1}{1 + e^{-\boldsymbol{\theta}^\top \boldsymbol{x}_i}}$$

Lets write this down in terms of the linear feature $\boldsymbol{\theta}^\top \boldsymbol{x}_i$.

$$\pi_i = \frac{e^{\boldsymbol{\theta}^\top \boldsymbol{x}_i}}{1 + e^{\boldsymbol{\theta}^\top \boldsymbol{x}_i}}$$

$$1 - \pi_i = \frac{1}{1 + e^{\boldsymbol{\theta}^\top \boldsymbol{x}_i}}$$

$$\Rightarrow$$

$$\frac{\pi_i}{1 - \pi_i} = e^{\boldsymbol{\theta}^\top \boldsymbol{x}_i}$$

$$\boldsymbol{\theta}^\top \boldsymbol{x}_i = \log \frac{\pi_i}{1 - \pi_i}$$

Logistic Regression — Maximum Likelihood Objective

Let $\boldsymbol{y} = [y_1, \ldots, y_n]^\top$ and $\boldsymbol{\pi} = [\pi_1, \ldots, \pi_n]^\top$.

*Use Maximum Likelihood formulation to find the $\boldsymbol{\theta}$ that **best predict** the label y_i.*

$$L(\boldsymbol{y} \mid \boldsymbol{\pi}) \sim \mathrm{Bern}(\boldsymbol{\pi})$$

$$= \prod_{i=1}^{n} \pi_i^{y_i}(1-\pi_i)^{1-y_i} \quad = \prod_{i=1}^{n} \left(\frac{\pi_i}{1-\pi_i}\right)^{y_i}(1-\pi_i)$$

$$= \prod_{i=1}^{n} e^{y_i \log\left(\frac{\pi_i}{1-\pi_i}\right) + \log(1-\pi_i)}$$

$$= \prod_{i=1}^{n} e^{y_i \boldsymbol{\theta}^\top \boldsymbol{x}_i - \log(1+e^{\boldsymbol{\theta}^\top \boldsymbol{x}_i})}$$

Logistic Regression objective function.

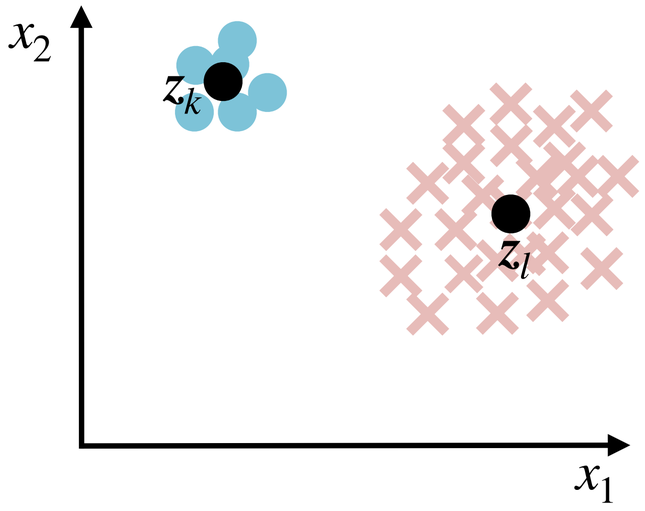
Cross-entropy.

How to Incorporate Prior Knowledge in Your Classification

What if one class is much less likely to happen?

Example:

$$p(y = k) \leq p(y = l)$$



We will use *Bayes' Rule* and *Posterior* to incorporate *Priors* into *Likelihood*.

Bayesian 2-Class Classification

Lets remind ourselves on WED decision criteria on the board.

Let $\{y_i \in \{0,1\}, \boldsymbol{x}_i \in \mathbb{R}^p\}_{i=1}{}^n$. Assume

$$p(\boldsymbol{x} \,|\, y = c) = \mathcal{N}(\bar{\boldsymbol{x}}^c, \boldsymbol{\Sigma})$$

Goal: predict $p(y_i \,|\, \boldsymbol{x}_i)$.

Bayes' Rule:

$$\overbrace{p(y = c \mid \boldsymbol{x})}^{\textit{posterior}} = \frac{\overbrace{p(\boldsymbol{x} \,|\, y = c)}^{\textit{likelihood}}\overbrace{p(y = c)}^{\textit{prior}}}{p(\boldsymbol{x})}$$

Assign examples to the class with the **Maximum a Posteriori (MAP)** rule.

$$\boldsymbol{x}_i \in c_k \text{ iff } p(y = c \mid \boldsymbol{x}) > p(y = l \,|\, \boldsymbol{x})$$

for all $l \neq k$

MAP Decision Boundary

For prediction $\hat{c} = 1$.

$$p(y = 1 \mid \boldsymbol{x}) \geq p(y = 0 \mid \boldsymbol{x})$$

$$\frac{p(y = 1 \mid \boldsymbol{x})}{p(y = 0 \mid \boldsymbol{x})} \geq 1$$

$$\frac{p(\boldsymbol{x} \mid y = 1)p(y = 1)}{p(\boldsymbol{x} \mid y = 0)p(y = 0)} \geq 1$$

(take the log)

$$\underbrace{\log p(\boldsymbol{x} \mid y = 1) - \log p(\boldsymbol{x} \mid y = 0)}_{\text{This is Likelihood from WED.}} + \underbrace{\log \frac{\pi_1}{\pi_0}}_{\text{This is from prior.}} \geq 0$$

MAP Decision Boundary (II)

$$p(\boldsymbol{x}\,|\,y=c) = \frac{1}{(2\pi)^{p/2}\,|\,\boldsymbol{\Sigma}\,|^{1/2}} e^{\frac{-1}{2}(\boldsymbol{x}-\bar{\boldsymbol{x}}^c)\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\bar{\boldsymbol{x}}^c)}$$

Following steps similar to MED and WED, we get

$$\underbrace{(\bar{\boldsymbol{x}}^1 - \bar{\boldsymbol{x}}^0)^\top \boldsymbol{\Sigma}^{-1}}_{\boldsymbol{\theta}_1^\top}\boldsymbol{x} \;-\; \underbrace{\tfrac{1}{2}({\boldsymbol{x}^1}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{z}^1 - {z_0}^\top \boldsymbol{\Sigma}^{-1} z_0) + \log\frac{\pi_1}{\pi_0}}_{\theta_0} \geq 0$$

This is similar to the decision rule from WED plus the bias term $\log\dfrac{\pi_1}{\pi_0}$ which takes into consideration how likely it is for you to get a data example from class 1 compared to class 0. Is your dataset balanced? With equal probability? This is your *prior* about how likely it is for a data to be from one class or another.

MAP Decision Rule vs. Logistic Regression

MAP decision rule:

$$\log \frac{p(y=1 \mid \boldsymbol{x})}{p(y=0 \mid \boldsymbol{x})} = \frac{p(y=1 \mid \boldsymbol{x})}{1-p(y=1 \mid \boldsymbol{x})} = \boldsymbol{\theta}^\top \boldsymbol{x} + \theta_0$$

Log-odds are linear in the feature with assumption Gaussian assumption of different mean, same covariance.

This is similar to the sigmoid in logistic regression:

$$\log \frac{\pi_i}{1-\pi_i} = \boldsymbol{\theta}^\top \boldsymbol{x}_i$$

Differences

Bayesian	**Logistic**
depends on $\pi_0, \pi_1, \boldsymbol{\Sigma}, \bar{\boldsymbol{x}}^0, \bar{\boldsymbol{x}}^1$	*Estimate directly the vectors using* π_0, π_1
$O(p^2)$	$O(p)$

Takeaways

- *Learning methods construct a decision criterion based on their assumption about data. They may drastically fail when their underlying data assumption is wrong.*

- *Lots of methods does Maximum Likelihood (ML) estimation.*

- *Logistic Regression: Maximum likelihood for Bernoulli distribution.*

- *Incorporate prior: Maximum a Posteriori (MAP) for Bayesian Classification.*